CHAPTER 02

Averages

Exams, such as CAT, XAT, IIFT, CMAT, GMAT, SSC CGL and Bank PO ask direct problems from this chapter. Looking at the historical data, we can say that every year they usually ask at least 1 problem in Quantitative Aptitude section and at least 4 problems in Data Interpretation section. Sometimes they ask a few questions which are application based and involve a great amount of logical thinking. It suggests that we need to develop a fresh logical approach to answer the problems of this chapter. Essentially, I won't recommend you to completely rely on formula-based approach.

Chapter Checklist

- Average
  - Properties of Average
  - Weighted Average
  - Problems based on Age
  - Problems based on Income and Salary
  - Problems based on Time, Speed and Distance
  - Value of an Overlapping Element
  - Average of Same Important Series of Numbers
- CAT Test

2.1 Average

In general average is the central value of the given data. For example, if the heights of three persons A, B and C be 90 cm, 110 cm and 115 cm, respectively, then the average height of A, B and C together will be $\frac{90+110+115}{3}=105$ cm. So, we can say that the height of each person *viz.* A, B and C is near about 105 cm. Thus in layman's language it can be said that every one is almost 105 cm tall. Basically, the average is the arithmetic mean of the given data. For example, if the $x_1, x_2, x_3, x_4 \ldots x_n$ be any 'n' quantities (*i.e.*, data), then the average (or arithmetic mean) of these 'n' quantities $= \frac{x_1 + x_2 + x_3 + \ldots + x_n}{n}$

Properties of Average

1. The average of any two or more quantities (or data) necessarily lies between the lowest and the highest values of the given data. *i.e.*, if x_l and x_h be the lowest and highest (or greatest) values of the given data $(x_1, x_2, \ldots x_l, \ldots x_h, \ldots, x_n)$ then $x_l < \text{Average} < x_h$; $x_l \neq x_h$ *i.e.* $x_l < \frac{(x_1 + x_2 + x_3 + x_l \ldots + x_h \ldots + x_n)}{n} < x_h$

Ex. 1) The average of 8, 9, 12, 13, 15, 9 is :

(a) 11 (b) 6 (c) 16 (d) 18

Solution From the above mentioned property (1) we can say that options (b), (c) and (d) are invalid since 6, 16 and 18 are out of range *i.e.*, either below 8 (which is the least value) or above 15 (which is the highest value of the data). So, only option (a) can be acceptable.

Alternatively Average $= \frac{8+9+12+13+15+9}{6} = \frac{66}{6} = 11$

Ex. 2) A has 8 pencils, B has 10 pencils and C has 15 pencils, then the average number of pencils with them :

(a) 8 (b) 10

(c) 15 (d) lies between 9 and 15

Solution Average number of pencils $= \frac{8+10+15}{3} = 11$

So, option (d) is correct.

2. If each quantity is increased by a certain value 'K' then the new average is increased by K.

Ex. 3) A, B, C, D and E are the five electronic shops in the Naza market, which have 20, 30, 60, 80 and 50. T.V. sets with them respectively, then the average number of T.V. sets in each shop is :

(a) 24 (b) 48 (c) 50 (d) 60

Solution Average number of T.V. sets $= \frac{20+30+60+80+50}{5} = 48$

So, option (b) is correct.

Ex. 4) In the previous example, if each of A, B, C, D and E have imported 12 new T.V. sets, then the average number of T.V. sets is:

(a) 50 (b) 56 (c) 60 (d) 144

Solution Average number of T.V. sets

$$= \frac{(20+12)+(30+12)+(60+12)+(80+12)+(50+12)}{5} = \frac{300}{5} = 60$$

Alternatively

$$= \frac{(20+30+60+80+50)+(12+12+12+12+12)}{5}$$

$$= \frac{(240)+(5\times 12)}{5} = 48+12 = 60$$

Thus, the new average will be 60.

Ex. 5) The average number of shirts with Salman, Ambani and Dalmiya is 60, if all of them reached a shopping mall (AMBI) in Gurgaon and purchased 6 shirts each of them then the average number of shirts each of them now has :

(a) 66 (b) 63

(c) 62 (d) can't be determined

Solution Required average = Old average + New average $= 60+6 = 66$

3. If each quantity is decreased by a certain value K, then the new average is also decreased by K.

Ex. 6) A, B, C, D, E and F are the only six families in Indira nagar. A, B, C, D, E and F has 7, 8, 10, 13, 6 and 10 members in their families respectively. If 1 member from all the six families left their respective families to accomodate themselves in the hostel of IIM Lucknow, then the average number of members now in each family of Indira nagar is :

(a) 8 (b) 9 (c) 10 (d) 13

Solution Required average

$$= \frac{(7-1)+(8-1)+(10-1)+(13-1)+(6-1)+(10-1)}{6}$$

$$= \frac{(7+8+10+13+6+10)}{6} - \frac{(6\times 1)}{6} = 9-1 = 8$$

Ex. 7) Last year Sahara, Tata, Singhania and Birla each has 250 industries. This year everyone has sold out 10 factories due to the recession of the economy and poor turnout. The average number of industries which each of them now has :

(a) 260 (b) 240

(c) 25 (d) none of these

Solution Required average = Old average – New average

$= (250) - (10) = 240$

4. If each quantity is multiplied by a certain value K, then the new average is the product of old average with K.

Ex. 8) In a flower shop there were 6 flowers in each bouquet. If the seller has doubled the number of flowers in each bouquet then the new average of flowers in each bouquet is :

(a) 12 (b) 8

(c) 3 (d) none of these

Solution Now required average

= Old average × Multiplicand $= 6\times 2 = 12$

5. If each quantity is divided by a certain quantity 'K', then the new average becomes $\frac{1}{K}$ times of the initial average, where $K \neq 0$.

Ex. 9) The average of 100, 200, 300, 400, …, 1000 is 550. If each number is divided by 5, then the new average will be equal to :

(a) 450

(b) 45

(c) 55

(d) none of the above

Solution Actual average $= \frac{(100+200+300+\ldots+1000)}{10} = 550$

The new average $= \frac{550}{5} = 110$

Hence, (d) is correct.

6. If 'A' be the average of $x_1, x_2, x_m, \ldots\ y_1, y_2, \ldots, y_n$. where $x_1, x_2, \ldots, x_m$ be the below A and $y_1, y_2, y_3, \ldots, y_n$ be the above A, then

$$(A-x_1)+(A-x_2)+\ldots(A-x_m)$$

$$= (y_1-A)+(y_2-A)+\ldots(y_n-A)$$

i.e., the surplus above the average is always equal the net deficit below average.

Ex. 10) The salary of A, B, C, D and E is ₹ 8000, ₹ 5000, ₹ 11000, ₹ 7000, ₹ 9000 per month respectively, then the average salary of A, B, C, D and E per month is :

(a) ₹ 7000 (b) ₹ 8000
(c) ₹ 8500 (d) ₹ 9000

Solution Average salary

$$= \frac{8000 + 5000 + 11000 + 7000 + 9000}{5} = ₹\ 8000$$

NOTE The average salary = ₹ 8000 and the sum of salaries above ₹ 8000 = 11000 + 9000 = 20000

Therefore, the average surplus $= \left(\frac{20000}{2}\right) - 8000 = 2000$

again, the sum of salaries below ₹ 8000 = 5000 + 7000 = 12000

Therefore, the average deficit $= 8000 - \left(\frac{12000}{2}\right) = 2000$

Hence, the surplus above the average = deficit below the average = 2000

Alternatively Let the average be 10000

Then,

$10000 - 8000 = 2000$
$10000 - 5000 = 5000$
$10000 - 7000 = 3000$ and $11000 - 10000 = 1000$
$10000 - 9000 = 1000$ Surplus = $\underline{1000}$
Deficit → $\underline{11000}$

$\therefore$ Net difference = Surplus – deficit
$= 1000 - 11000 = -10000$

$\therefore$ Average variation $= \frac{-10000}{5} = -2000$

Hence, the actual average = assumed average + average variation
$= 10000 - 2000 = 8000$

Ex. 11) The average of 6 numbers 13, 17, 25, 11, 26, 10 is:

(a) 20 (b) 17 (c) 15 (d) 25

Solution Let the average be 20.

Then,
$20 - 13 = 7$ $20 - 25 = 5$
$20 - 17 = 3$ and $20 - 26 = 6$
$20 - 11 = 9$ $\underline{11}$
$20 - 10 = 10$ (Surplus)
$\underline{29}$ (deficit)

The net difference (surplus – deficit) = (11 – 29) = –18

Therefore average variation $= \frac{-18}{6} = -3$

Hence, the actual average = 20 + (–3) = 17

NOTE This method is very helpful in Data Interpretation section where there are very large values are to be calculated and most of the time we need just the lumpsum value.

7. Weighted Average : When the average of groups or sets, instead of individuals, having different number of elements is being calculated, then it is called the weighted average. Since in this case the number of elements is different for the different sets thus they carry different weightage. If the number of elements in n different groups be $K_1, K_2, K_3, K_4, \ldots, K_n$ and the averages of the respective groups be $A_1, A_2, A_3, A_4 \ldots A_n$ then the

$$\text{weighted average} = \frac{K_1A_1 + K_2A_2 + K_3A_3 + \ldots K_nA_n}{K_1 + K_2 + K_3 + \ldots K_n}$$

Ex. 12) The average salary of 12 employees of STAR plus is ₹ 18000 per month and 15 employees of NDTV is ₹ 16000 per month. The average salary of all the 27 employees is :

(a) ₹ 17000 (b) ₹ 16500
(c) ₹ 16888.88 (d) none of these

Solution Required average salary

$$= \frac{12 \times 18000 + 15 \times 16000}{12 + 15} = \frac{456000}{27} = ₹\ 16888.88$$

Ex. 13) The number of students at $i4$ IIM in morning batch, evening batch and weekends batch is 30, 40 and 60 respectively and their respective averages ages (in years) are 22, 21 and 25, then the average age of all the students (combined) is :

(a) $22\frac{1}{13}$ years (b) $23\frac{1}{13}$ years
(c) 24.5 years (d) none of these

Solution $\frac{30 \times 22 + 40 \times 21 + 60 \times 25}{130} = \frac{3000}{130} = 23\frac{1}{13}$ years

8. Problems Based on Age :

(i) If the average age of 'n' members of a family is x years then K years back, the average age of the family is $(x - K)$ years, provided that no any person is expired or born in K years period.

(ii) If the present age of 'n' members of a family is x years then K years later the average age of the same family will be $(x + K)$ years.

Ex. 14) The average age of 7 members of Patel's family is 25 years. The average age of the same family 3 years ago was :

(a) 21 years (b) 22 years
(c) 25 years (d) none of these

Solution Present average age of family = 25 years

3 years ago average age of family = 25 – 3 = 22 years

Ex. 15) The average age of all the 16 professors of Lucknow University is 52 years. Four years later the average age of all the 16 professors will be, if there is no any retirement or recruitment :

(a) 68 (b) 64
(c) 56 (d) none of these

Solution Present average age = 52 years

4 years later the average age = 52 + 4 = 56 years

Ex. 16) 6 months ago the present age of the student of class 10th was 14 years. 6 months hence, the age of the same students will be :

(a) 15 years (b) $15\frac{1}{2}$ years
(c) 20 years (d) none of these

Solution Since the time difference between two dates is 1 year, hence the average age will be increased by 1 year. Thus the average age of the class 6 months hence will be 15 years.

Ex. 17) The average age of Priyambada's family consisting of 5 members 3 years ago was 35 years. One year ago a new baby was born in this family. Three years hence the average age of the family will be :

(a) 36 years (b) $34\frac{5}{6}$ years
(c) $35\frac{4}{5}$ years (d) none of these

Solution 3 years ago total age of 5 members

$= 5 \times 35 = 175$ years

At the time of birth of new baby the total age of family

$= 175 + (2 \times 5) = 185$ years

The present age of family $= 185 + (1 \times 6) = 191$ years

3 years hence, the average age of family

$$= \frac{191 + (3 \times 6)}{6} = 34\frac{5}{6} \text{ years}$$

Ex. 18) 10 years ago the average age of all the 25 teachers of the Girls college was 45 years. 4 years ago, the principal has retired from her post at the age of 60 year. So after one year a new principal whose age was 54 years recruited from outside. The present average age of all the teachers is, if principal is also considered as a teacher :

(a) $54\frac{18}{25}$ years (b) $55\frac{17}{25}$ years
(c) $49\frac{1}{2}$ years (d) none of these

Solution 10 years ago average age of 25 teachers = 45 years

4 years ago (just before the retirement of principal) average age of 25 teachers = 45 + 6 = 51 years

and the same time total age of 25 teachers $= 51 \times 25$

$= 1275$ years

and the total age of remaining 24 teachers when just the principal has retired $= 1275 - 60 = 1215$ years

1 year later (*i.e.*, 3 years ago from present) total age of 24 teachers (just before the recruitment of new principal)

$= 1215 + (1 \times 24) = 1239$ years

and the total age of 25 teachers including new principal just after the recruitment

$= 1239 + 54 = 1293$ years

Thus the present age of all the 25 teachers

$= 1293 + (3 \times 25) = 1368$ years

Hence, the present average age of the 25 teachers

$$= \frac{1368}{25} = 54\frac{18}{25} \text{ years}$$

Alternatively 10 years ago, the average age of 25 teachers = 45 years

Let us assume that the principal has not retired from her post then the present average age of all the 25 teachers,

$= 45 + 10 = 55$ years

Thus the total age of all the 25 teachers

$= 55 \times 25 = 1375$ years

Now, assume that the new principal has replaced the old principal 4 years ago instead of 3 years ago, when the age of new principal would have been 53 years. Thus the age of new principal was 7 years less than the age of old principal, which results in the reduction of total age of the group of 25 teachers by 7 years.

Thus the actual total age of the 25 teachers (presently)

$= 1375 - 7 = 1368$ years

Hence, the present average age of the 25 teachers

$$= \frac{1368}{25} = 54\frac{18}{25} \text{ years}$$

Ex. 19) The ratio of the ages of the father and the daughter at present is 3 : 1. Four years ago the ratio was 4 : 1. The average age of the father and daughter 2 years hence will be :

(a) 24 (b) 26 (c) 25 (d) 36

Solution Let the present ages of father be $3x$ and daughter be x. So the 4 years ago father's age and daughter's age was $(3x - 4)$ and $(x - 4)$. Therefore $\left(\frac{3x-4}{x-4}\right) = \frac{4}{1}$

$\Rightarrow$ $x = 12$ years and $3x = 36$ years

Hence, the present average age of father and daughter = 24 years and the average age 2 years hence will be 26 years.

9. Problems Based on Income/Salary

Income = Expenditure + Savings

Ex. 20) The average salary of Rajesh, Bahadur and Amir is ₹ 8000 per month. The average expenditure of the Rajesh, Bahadur and Amir per month is ₹ 5000. The average savings of all the 3 persons per month is :

(a) ₹ 3000 (b) ₹ 5000 (c) ₹ 2500 (d) ₹ 9000

Solution Average saving

= Average Income – Average Expenditure

= 8000 – 5000 = 3000

Ex. 21) The average salary of A, B and C is ₹ 10000 and average expenditure of A is ₹ 6000 then the average savings of B and C is :

(a) ₹ 5500 (b) ₹ 4500
(c) ₹ 4000 (d) can't be determined

Solution Total income = Total Expenditure + Total savings. We can not find the average savings of B and C, since data is insufficient.

Ex. 22) The average salary of A, B is ₹ 6000 and that of C, D and E is ₹ 8000. The average salary of all the 5 people is

(a) ₹ 7200 (b) ₹ 7000
(c) ₹ 7500 (d) can't be determined

Solution Required average salary

$$= \frac{6000 \times 2 + 8000 \times 3}{(2+3)} = \frac{36000}{5} = ₹ 7200$$

Ex. 23) The average salary of all the 60 employees in an office is ₹ 12,000 per month. If the number of executives is twice the number of non-executive employees, then the average salary of all the non executive employees is :

(a) ₹ 9000 (b) ₹ 8000
(c) ₹ 6000 (d) can't be determined

Solution $\frac{\text{No. of executives}}{\text{No. of non-executives}} = \frac{2}{1}$

Therefore number of executives = 40
and number of non-executive employees = 20
Now, go through the options
Total salary = 40 × salary of executive + 20 × salary of non-executive
$60 \times 12000 = 40 \times k + 20 \times l$, here k, l are unknowns
So we can't determine the required average salary.

Ex. 24) In the above problem, if the average salary of non-executives be 2/5th of the average salary of executives, then the average salary of non-executive employees is :

(a) ₹ 9000 (b) ₹ 8000
(c) ₹ 6000 (d) data insufficient

Solution By options : $60 \times 12000 = 20 \times 6000 + 40 \times 15000$
Hence option (c) is correct.

Alternatively $60 \times 12000 = 20 \times x + 40 \times \frac{5}{2} x \Rightarrow x = 6000$

10. Problems Based on Time, Speed and Distance :

Case 1. When the distance travelled in different time slots or parts is same *i.e.*, if a person or vehicle moves x km at a speed of u km/hr and further he goes or comes back the same distance x km at a speed of v km/hr.

Then the average speed $= \frac{2uv}{(u+v)}$. If there are 3 parts of distance x km travelled with 3 different speeds *i.e.*, if a person goes first x km @ speed of u km/hr and next x km @ v km/hr and the last x km @ w km/hr.

Then the average speed $= \frac{3uvw}{(uv+vw+wu)}$.

NOTE In this type of questions the average speed is independent of the distance travelled.

General Formula : Average speed $= \frac{\text{Total distance}}{\text{Total time}}$

The above two formulae are derived with the help of the general formula of average speed.

Proportion Method : If the half of the distance is covered at u km/hr and the rest half of the journey is covered at v km/hr then the average speed can be found as follows :

***Step* 1.** Divide the difference of u and v in the ratio of $u : v$ (where $u < v$)

***Step* 2.** $u + \frac{(u \sim v) \times u}{(u+v)}$ or $v - \frac{(u \sim v) \times v}{u+v}$

Case 2. When the distances travelled at different speeds are different then we calculate the average speed with the help of general formula of average speed. *e.g.*, A person first goes x_1 km at the speed of u km/hr and x_2 km at the speed of v km/hr and x_3 km at the speed of w km/hr and so on, then the

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}} = \frac{x_1 + x_2 + x_3 + \ldots}{t_1 + t_2 + t_3 + \ldots}$$

$$= \frac{x_1 + x_2 + x_3 + \ldots}{\frac{x_1}{u} + \frac{x_2}{v} + \frac{x_3}{w} + \ldots}$$

Ex. 25) Einstien goes from Ahmedabad to Lucknow at the speed of 40 km/hr and returned at the speed of 60 km/hr. The average speed of Einstien during the whole journey is :

(a) 48 km/hr
(b) 24 km/hr
(c) 50 km/hr
(d) none of the above

Solution $\frac{2 \times 40 \times 60}{(40+60)} = \frac{4800}{100} = 48$ km/hr

So the average speed $= 40 + \frac{2}{5}(20) = 48$ km/hr

or Average Speed $= 60 - \frac{3}{5}(20) = 48$ km/hr

Ex. 26) Sri Krishna took the chariot and started his journey from Mathura to Gokul by his chariot at the speed of 40 km/hr and then, the same distance he travelled on his foot at the speed of 10 km/hr from Gokul to Brindaban. Then he returned from Brindaban to Mathura via Gokul at the speed of 24 km/hr riding on the horse. The average speed of the whole trip is :

(a) 20 km/hr
(b) 25 km/hr
(c) 19.2 km/hr
(d) 18.5 km/hr

Solution Since the distance from Mathura to Gokul is same as that of Gokul to Brindaban. So the average speed from Mathura to Brindaban $= \frac{2 \times 40 \times 10}{(40 + 10)} = 16 \text{ km/hr}$

Again since he returned on the same path, so the distance from Mathura to Brindaban is same in both the directions. Thus the required average speed $= \frac{2 \times 16 \times 24}{(16 + 24)} = 19.2 \text{ km/hr}$

Ex. 27) Anoop travels first $\frac{1}{3}$rd of the total distance at the speed of 10 km/hr and the next $\frac{1}{3}$rd distance at the speed of 20 km/hr and the last $\frac{1}{3}$rd distance at the speed of 60 km/hr. The average speed of Anoop is :

(a) 15 km/hr (b) 18 km/hr
(c) 25 km/hr (d) 30 km/hr

Solution Since all the three distances are same, hence the average speed

$$= \frac{3 \times 10 \times 20 \times 60}{(200 + 1200 + 600)} \qquad \left(= \frac{3uvw}{uv + vw + wu}\right)$$

$$= \frac{36000}{2000} = 18 \text{ km/hr}$$

Ex. 28) Columbus started his journey from Lucknow to Kolkata, which is 200 km, at the speed of 40 km/h then he went to Bangalore which is 300 km, at the speed of 20 km/hr. Further he went to Ahmedabad which is 500 km, at the speed of 10 km/hr. The average speed of Columbus is :

(a) $14\frac{2}{7}$ km/hr (b) $14\frac{5}{7}$ km/hr
(c) 15.6 km/hr (d) none of these

Solution $\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}} = \frac{(200 + 300 + 500)}{\left(\frac{200}{40} + \frac{300}{20} + \frac{500}{10}\right)}$

$$= \frac{1000}{70} = \frac{100}{7} = 14\frac{2}{7} \text{ km/hr} = 14.2857 \text{ km/hr}$$

11. Value of An Overlapping Element

(i) $a + b = k$ and $b + c = l$ and $a + b + c = m$
then $[(a + b) + (b + c)] - (a + b + c) = (k + l) - m$
or $b = k + l - m$

(ii) $a + b = k$, $d + e = l$ and $a + b + c + d + e = m$
then $c = (a + b + c + d + e) - [(a + b) + (d + e)] = m - [k + l]$

Ex. 29) The average weight of all the 11 players of Indian cricket team is 50 kg. If the average of first six lightest weight players is 49 kg and that of the six heaviest players is 52 kg. The average weight of the player which lies in the sixth position in the list of players when all the 11 players are arranged in the order of increasing or decreasing weights :

(a) 56 kg (b) 52 kg
(c) 51 kg (d) none of these

Solution Let A, B, C, D, E, F, G, H, I, J and K be the 11 players in the order of increasing weight then,

$$A + B + C + D + E + F = 49 \times 6 = 294$$
$$F + G + H + I + J + K = 52 \times 6 = 312$$
and $A + B + C + D + E + F + G + H + I + J + K = 50 \times 11 = 550$
Therefore $F = (A + B + C + D + E + F) + (F + G + H + I + J + K)$
$$- (A + B + C + D + \ldots J + K) = 294 + 312 - 550 = 56$$
Hence the average weight of $F = 56$ kg.

Ex. 30) The average presence of students in a class on Monday, Tuesday and Wednesday is 30 and on the Wednesday, Thursday, Friday and Saturday is 28 then number of students who attended the class on Wednesday is, if the average number of students on all the six days is 27

(a) 24 (b) 25 (c) 20 (d) 40

Solution Since $W = (M + T + W) + (W + Th + F + S)$
$$- (M + T + W + Th + F + S)$$
$$= (30 \times 3) + (28 \times 4) - (27 \times 6) = 202 - 162 = 40$$

Ex. 31) The average age of A, B, C, D and E is 40 years. The average age of A and B is 35 years and the average age of C and D is 42 years. The average age of E is :

(a) 46 (b) 48
(c) 32 (d) none of these

Solution $A + B + C + D + E = 40 \times 5 = 200$
$$A + B = 35 \times 2 = 70; C + D = 42 \times 2 = 84$$
Therefore, $E = (A + B + C + D + E) - [(A + B) + (C + D)]$
$$= 200 - (70 + 84) = 46$$
Thus the average age of $E = 46$ years.

Ex. 32) The average temperature on Monday, Tuesday and Wednesday is 38°C. The average temperature on Tuesday, Wednesday and Thursday is 43°C. If the average temperature on Monday and Thursday is 18.5°C. The average temperature on Monday is :

(a) 11°C (b) 21°C (c) 35°C (d) 27°C

Solution
$(M + T + W) = 38 \times 3 = 114$ …(i)
$(T + W + Th) = 43 \times 3 = 129$ …(ii)
Therefore, $Th - M = 15$ [(ii) – (i)]
and $Th + M = 37$
Thus the temperature on Monday is 11°C and on Thursday is 26°C.

12. Average of Some Important Series of Numbers :

(i) Average of first 'n' natural numbers $= \frac{n + 1}{2}$

(ii) Average of first 'n' even numbers $= (n + 1)$

(iii) Average of first 'n' odd numbers $= n$

(iv) If there are $(p + q)$ elements in a set or group but the average of p elements is r and the average of q elements is s, then the average of all the elements of the set (or group) is $\frac{(pr + sq)}{(p + q)}$.

CAT-Test

Questions Helping you bell the CAT

LEVEL 01 > BASIC LEVEL EXERCISE

1 The average weight of a class of 20 students is 45 kgs. A new student whose weight is 40 kgs replaces an old student of this class. Hence, the average weight of the whole class decreases by 1 kg. The weight of the replaced student is :
(a) 55 kgs (b) 50 kgs
(c) 60 kgs (d) none of these

2 The average length of first 3 fingers is 3 inches and the average length of the other 2 fingers *i. e.*, thumb and the index finger is 2.8 inches. If the length of the index fingers is 3 inches then the length of the thumb is :
(a) 2 inches (b) 2.6 inches
(c) 3 inches (d) none of these

3 Three types of rice whose rates are ₹ 38, ₹ 43 and ₹ 49 per kg are blended together to make a 15 kg of new blend of rice in which there are 8 kgs, 4 kgs and 3 kgs of the respective types of rice. The average price of the new blend of rice is :
(a) ₹ 41.53 (b) ₹ 43
(c) ₹ 40 (d) ₹ 43.3

4 Pankaj went to the post-office at the speed of 60 km/hr while returning for his home he covered the half of the distance at the speed of 10 km/hr, but suddenly he realized that he was getting late so he increased the speed and reached the home by covering rest half of the distance at the speed of 30 km/hr. The average speed of the Pankaj in the whole length of journey is :
(a) 5.67 km/hr
(b) 24 km/hr
(c) 22.88 km/hr
(d) 5.45 km/hr

5 123 typists typed 984 papers in 1/15 hour. The number of papers typed per minute by an average typist is :
(a) 1 (b) 2
(c) 3 (d) 5

6 The cost of the Red, Green and Blue colours per kg is ₹ 15 and ₹ 18 respectively. Rang Mahal is a renowned building in which these three colours are being used in the ratio of 3 : 2 : 4. The average cost of all the three colours used per kg is :
(a) 18 (b) 20
(c) 17.66 (d) can't be determined

7 Rajeev earns 3/2 times in January, April, July and October than his average earning of ₹ 600 per month in the rest of the months. Therefore, his savings in the month of January, April, July and October is 5/4 times of the monthly savings of ₹ 400 for each of the remaining months of the year. The average expenditure of per month is :
(a) ₹ 266.66 (b) ₹ 250 (c) ₹ 233.33 (d) ₹ 433.33

8 The average of 3 prime numbers lying between 47 and 74 is 191/3. The greatest possible difference between any two out of the 3 prime numbers is :
(a) 12 (b) 14
(c) 18 (d) can't be determined

9 The average of 9 numbers is 11. If each of these 9 numbers is multiplied by 5 and then 5 is added to each of these resultant numbers, then the new average is :
(a) 20 (b) 30 (c) 60 (d) 50

10 The average score of Dhoni after 48 innings is 48 and in the 49th innings Dhoni scores 97 runs. In the 50th innings the minimum number of runs required to increase his average score by 2 than it was before the 50th innings :
(a) 99 (b) 149
(c) 151 (d) can't be determined

11 The average age of Sachin and Ganguli is 35 years. If Kaif replaces Sachin, the average age becomes 32 years and if Kaif replaces Ganguli, then the average age becomes 38 years. If the average age of Dhoni and Irfan be half of the average age of Sachin, Ganguli and Kaif, then the average age of all the five people is :
(a) 28 years (b) 32 years (c) 25 years (d) none

12 Out of these five people (in question no. 11) whose age is the greatest?
(a) Sachin (b) Ganguli
(c) Kaif (d) can't be determined

13 In a village the average age of *n* people is 42 years. But after the verification it was found that the age of a person had been considered 20 years less than the actual age, so the new average, after the correction, increased by 1. The value of *n* is :
(a) 21 (b) 20
(c) 22 (d) none of these

14 The average rainfall in the months of January and February is 6 cm and in the months of March to June is 5 cm and July to October is 10 cm and in the November and December, it is 6 cm. The average rainfall for the whole year is :
(a) 7 (b) 5.5
(c) 7.5 (d) none of these

15 On an average 300 people watch the movie in Sahu Cinema hall on Monday, Tuesday and Wednesday and the average number of visitors on Thursday and Friday is 250. If the average number of visitors per day in the week be 400, then the average number of people who watch the movie in weekends (*i. e.*, on Saturday and Sunday) is :
(a) 500 (b) 600
(c) 700 (d) none of these

16 The average weight of 11 players of Indian cricket team is increased by 1 kg, when one player of the team weighing 55 kg replaced by a new player. The weight of the new player is :
(a) 55 kg (b) 64 kg
(c) 66 kg (d) none of these

17 The average age of a family of 6 members 4 year ago was 25 years. Mean while a child was born in this family and still the average age of the whole family is same today. The present age of the child is :
(a) 2 years (b) $1\frac{1}{2}$ years
(c) 1 year (d) data insufficient

18 Amitabh's average expenditure for the January to June is ₹ 4200 and he spends ₹ 1200 in January and ₹ 1500 in July. The average expenditure for the months of February to July is :
(a) ₹ 4250 (b) ₹ 4520
(c) ₹ 4060 (d) none of these

19 The average of *a*, 11, 23, 17 is 15 and the average of *a*, *b*, 12, 25 is 16. The value of $\frac{a}{b}$ is :
(a) 1/3 (b) 1/2 (c) 2/3 (d) 3/4

20 The average salary is being paid to all its employees by the Biotech corporation is ₹ 15500. The average salary of the senior employees is ₹ 18000 per month and the average salary of the junior employees is ₹ 12000 per month. If there are only two levels of employees viz junior and senior level, then what fraction of the total employees is the junior level employees are :
(a) $\frac{7}{10}$ (b) $\frac{5}{12}$
(c) 5/10 (d) none of these

21 The average of any 5 consecutive odd numbers *a*, *b*, *c*, *d* and *e* is :
(a) $\frac{(abcde)}{5}$ (b) $\frac{bd}{3}$
(c) $\left(\frac{a+c+e}{5}\right)$ (d) none of these

22 The average age of 30 students of a class is 30 years. When the average age of class teacher is also included, the average age of the whole class increases by 1 year. The age of the class teacher is :
(a) 31 years (b) 60 years
(c) 61 years (d) none of these

23 There were five sections in MAT paper. The average score of Pooja in first 3 sections was 83 and the average in the last 3 sections was 97 and the average of all the sections (*i. e.*, whole paper) was 92, then her score in the third section was :
(a) 85 (b) 92
(c) 88 (d) none of these

24 The average age of 18 pupils of Dronacharya was 25 years. If the age of Dronacharya was also included, the average age of 19 people becomes 26 years. The average age of the Dronacharya at that time was :
(a) 33 (b) 44
(c) 50 (d) 51

25 The average of 7 consecutive odd numbers if the smallest of those numbers is denoted by k :
(a) $k + 4$ (b) $k + 7$
(c) $k + 6$ (d) $7k$

26 If the average marks of $\frac{1}{4}$th class is 85% and that of $\frac{1}{3}$rd class is 70% and the average marks of the rest class is 56%, then the average of the whole class is (for the given subjects) :
(a) 67.916% (b) 72.33%
(c) 69.165% (d) can't be determined

27 The average length of any four fingers of my left hand is 600 mm. Then the average length of all the five fingers of my left hand is :
(a) 800 mm (b) 750 mm
(c) 480 mm (d) can't be determined

28 The average of $4\frac{3}{5}, 2\frac{2}{3}, 6\frac{8}{9}, 7\frac{7}{15}, 3\frac{5}{9}$ is :
(a) $5\frac{3}{225}$ (b) $5\frac{8}{225}$ (c) $6\frac{3}{45}$ (d) $25\frac{8}{45}$

29 The average of 1000.0001, 100.001, 10.01, 1.1 is :
(a) 277.777 (b) 322.222
(c) 11.11 (d) 233.333

30 The average of 7 consecutive numbers which are positive integers is 10. The average of lowest and highest such numbers is :
(a) 7 (b) 10
(c) 15 (d) data insufficient

31 The average of first 100 natural number is :
(a) 100 (b) 50
(c) 50.50 (d) 55

32 The average of first 50 odd natural numbers is :
(a) 50 (b) 55
(c) 51 (d) 101

33 The average of first 99 even numbers is :
(a) 9999 (b) 100
(c) 9801 (d) 9009

34 The average of a, b and c is 79 and the average of a and c is also 79. Then the value of b is :
(a) 0 (b) 79
(c) -79 (d) none of these

35 The average value of property of Mittal, Ambani and Singhania is ₹ 11111 crore. The property of Singhania is as less as the property of Mittal is greater than the average property of both the Singhania and Mittal. The value of property of Ambani is :
(a) ₹ 111 crore (b) ₹ 11111 crore
(c) ₹ 3703.7 crore (d) can't be determined

36 I went to Delhi @ speed of 200 km/hr but suddenly I returned to the same place @ speed of 600 km/hr. What is my average speed :
(a) 300 km/hr (b) 400 km/hr
(c) 366.66 km/hr (d) none of these

37 The average of A and B is 400 and the average of C and D is 600 the average of A, B, C and D is :
(a) 500 (b) 450
(c) 525 (d) 625

38 The average weight of liquid in 100 bottles is 500 gm. The total weight of all the bottles is 20 kg. The average weight of a bottle with liquid is :
(a) 0.65 kg (b) 0.7 kg
(c) 70 ml (d) none of these

39 The average score of Sehwag in 10 innings was 77 runs. In the 11th innings he had scored zero runs. The overall average score of Sehwag in all the 11 innings was :
(a) 77 (b) 7.7
(c) 11 (d) none of these

40 The average age of 3 children of Arihant Singh is 12 years and their ratio of ages is 3 : 4 : 5. The average age of the youngest and eldest child is if he had only 3 children :
(a) 12 (b) 21
(c) 8 (d) 9

41 The average income of all the Infosys employees is ₹ 20000 per month. Recently the company announced the increment of ₹ 2000 per month for all the employees. The new average of all the employees is :
(a) ₹ 22000
(b) ₹ 40000
(c) ₹ 2200
(d) data insufficient

42 The average age of 10 students in a class is 20 years, if a new student is also included, then the new average age of all the students increases by 1 year. The age of the new student is :
(a) 21 years (b) 30 years
(c) 31 years (d) none of these

43 The average weight of 20, four wheelers is 180 kg. If an old car is removed from this group of four wheelers, the new average weight decreases by 2 kg. The weight of the removed car is :
(a) 220 (b) 218
(c) 182 (d) none of these

44 The average price of 3 diamonds of same weights is ₹ 5 crore, where the average price of the two costliest diamonds is double the price of the cheapest diamond. The price of the cheapest diamond is :
(a) 3 crore (b) 5 crore
(c) 1.66 crore (d) can't be determined

45 In the previous question, the price of the costliest diamond is :
(a) 5 crore (b) 6 crore
(c) 8 crore (d) can't be determined

46 Praveen gets 40 marks out of 50 in Computer Science, 70 out of 100 in Manufacturing Science and 142 out of 150 in Professional communication. The average marks of Praveen (in percent) in all the three subjects is :
(a) 84% (b) 76% (c) 71% (d) 60%

47 The average of all the prime and composite numbers upto 100 is :
(a) 51 (b) 49.50
(c) 50.50 (d) 55

48 The average of all the perfect squares upto 100 is :
(a) 38.5 (b) 1000 (c) 100 (d) 385

49 The average of all the non-negative integers upto 99 is :
(a) 50.49 (b) 49.50
(c) 50.50 (d) 99

50 The average of 7, 14, 21, 28, … , 77 is :
(a) 7 (b) 11
(c) 42 (d) 66

51 The average weight of A, B, C and D is 40 kg. A new person E is also included in the group, then the average weight of the group is increased by 1 kg. Again a new person F replaces A, then the new average of 5 persons becomes 42. The average weight of B, C, D and F is :
(a) 42 (b) 41.25
(c) 42.5 (d) none of these

52 The average of 3 consecutive natural numbers (which are in increasing order) is k. If two more consecutive number, just next the first set of numbers, is added, then the new average becomes :
(a) $k+2$ (b) $k+1$ (c) $\frac{2k+1}{2}$ (d) $2k-1$

53 The average of any 5 consecutive odd natural numbers is k. If two more such numbers, just next to the previous 5 numbers are added, the new average becomes :
(a) $\frac{2}{7}(k+1)$ (b) $2k-3$
(c) $2k+1$ (d) $k+2$

54 The average weight of the 5 officers of a regiment is 42 kg. If a senior officer was replaced by a new officer and thus the average increased by 500 gm, the weight of the new officer is :
(a) 44.5 kg (b) 45 kg
(c) 42.5 kg (d) can't be determined

55 The average age of 6 servants in my farm house is 28 years. A new and young servant replaces an old servant, then the new average reduces by 1 year, the age of the new servant is :
(a) 26 years (b) 22 years
(c) 35 years (d) can't be determined

56 In the above question (no. 55) if the age of the replaced servant was 31 years, then the age of the new servant is :
(a) 25 years (b) 35 years
(c) 24 years (d) none of these

57 The average income of A, B and C is ₹ 12000 per month and the average income of B, C and D is ₹ 15000 per month. If the average salary of D be twice that of A, then the average salary of B and C is (in ₹) :
(a) 8000 (b) 18000
(c) 13500 (d) 9000

58 The average price of 80 computers in an electronic shop is ₹ 30000. If the highest and lowest price computers are sold out then the average price of the remaining 78 computers is ₹ 29500. The cost of the highest price computer is ₹ 80000. The cost of lowest price computer is
(a) ₹ 19000 (b) ₹ 20000
(c) ₹ 29000 (d) can't be determined

59 A has 50 coins of 10 paise denominations. While B has 10 coins of 50 paise denominations. C has 20 coins of 25 paise denominations while D has 25 coins of 20 paise denominations. The average number of paise per person is
(a) 450 paise (b) 500 paise
(c) 600 paise (d) can't be determined

60 A travel agency has three types of vehicles viz. four seater, autorickshaw, 10 seater maxi cab and 20 seater minibus. The rate of each passanger (irrespective of its age or weight or seniority) for the auto rickshaw is ₹ 12 and for the maxicab is ₹ 15 and for the minibus is ₹ 8 for the one round. The average occupancy of the seats is 100%, 80% and 75% respectively. If he has only one vehicle of each kind, then the average earning for one round of each vehicle is :
(a) ₹ 96 (b) ₹ 90 (c) ₹ 86 (d) ₹ 70

LEVEL 02 > HIGHER LEVEL EXERCISE

1 A train normally covers a certain distance at a speed of 60 km/hr. However, if it were to halt for a fixed time interval in each hour its average speed reduced to 50 km/hr. What is the time interval for which the train halt in each hour?
(a) 10 minutes (b) 20 minutes
(c) 6 minutes (d) 12 minutes

2 If p, q and r be three positive numbers such that $p > q > r$ when the smallest number is added to the difference of the rest two numbers, then the average of the resultant number and the original numbers except to the smallest number is 21 more than the average of all the three original numbers. The value of $(p - q)$ is :
(a) 7 (b) 14
(c) 63 (d) 42

3 Progressive express left for New Delhi, increasing its speed in each hour. It started its journey from Lucknow, but after four hours of its journey it met with accident. Its speed in the fourth hour was $\frac{7}{5}$ times that of the third hour and the speed in the third hour was $\frac{10}{7}$ times that of the second hour and in the second hour it was $\frac{7}{5}$ times that of the first hour. If it would have been travelled with the half of the speed that of the third hour, then it would have gone 160 km less in the same time (*i. e.*, in four hours). The average speed of the train during the journey of 4 hours was :
(a) 50 km/hr
(b) 90 km/hr
(c) 80 km/hr
(d) can't be determined

4 In a Mock CAT 123 students appeared and the average score obtained was 120. But later it was found that the top three students were repeaters, so their score has been eliminated and then the new average score was found to be decreased by 1.5. Also, it is known that all the students obtained the marks in integers and the scores of the toppers were distinct. If the second highest topper has scored more than 185 marks, then the highest possible score of the third highest topper is :
(a) 166 (b) 167
(c) 168 (d) 170

5 The average age of all the 100 employees in an office is 29 years, where $\frac{2}{5}$ employees are ladies and the ratio of average age of men to women is 5 : 7. The average age of female employees is :
(a) 18 years (b) 35 years
(c) 25 years (d) none of these

6 There are two houses in Parliament. One is Lok Sabha and the other one is Rajya Sabha and the member of Parliaments (MPs) in both the houses is 300 and 200 respectively. The average age of the members of Lok Sabha and Rajya Sabha is 40 years and 50 years respectively. A member of the Rajya Sabha when elected for the Lok Sabha also, he left the Rajya Sabha and becomes the member of the Lok Sabha. Thus the average age of both the houses increases. Which one of the following statements is true?
(a) The age of this member is greater than 50 years
(b) The age of this member is less than 40 years
(c) The age of this member is greater than 40 but less than 50 years
(d) none of the above

Directions (for Q. Nos. 7 to 9) *Eight years ago there were 5 members in the Arthur's family and then the average age of the family was 36 years. Mean while Arthur got married and gave birth to a child. Still the average age of his family is same now.*

7 The present age of his wife is :
(a) 25 years (b) 26 years
(c) 32 years (d) data insufficient

8 The age of his wife at the time of his child's birth was. If the difference between the age of her child and herself was 26 years :
(a) 25 years
(b) 26 years
(c) 20 years
(d) can't be determined

9 The age of Arthur at the time of his marriage was :
(a) 22 years (b) 23 years
(c) 26 years (d) can't be determined

10 Eleven years earlier the average age of a family of 4 members was 28 years. Now the age of the same family with six members is yet the same, even when 2 children were born in this period. If they belong to the same parents and the age of the first child at the time of the birth of the younger child was same as there were total family members just after the birth of the youngest members of this family, then the present age of the youngest member of the family is :
(a) 3 years (b) 5 years
(c) 6 years (d) none of these

11 Mr. Patel walked 6 km to reach the station from his house, then he boarded a train whose average speed was 60 km/hr and thus he reached his destination. In this way he took total of 3 hours. If the average speed of the entire journey was 32 km/hr then the average speed of walking is :
(a) 3 km/hr (b) 4.5 km/hr
(c) 4 km/hr (d) none of these

Directions (for Q. Nos. 12 and 13) *During the winter session all the women which constitutes* $\frac{1}{5}$*th strength of the Lok Sabha, left the house (i.e., parliament) due to the rejection of their demand. Actually they were asking for the* 50% *reservation of seats for the women in the Lok Sabha. Thus the average age of the remaining members of the house (i.e, the Lok Sabha) increases by* $\frac{1}{4}$*th than it was earlier when all the members (i.e., men and women) were present.*

12 The total strength of the Lok Sabha is :
(a) 300 MPs (b) 475 MPs
(c) 526 MPs (d) none of these

13 If the average age of women be 30 years then the average age of male members in the Lok Sabha will be :
(a) 26 (b) 34
(c) 40 (d) none of these

14 An artist in a circus moves along the hexagonal path of each side 20 metres in such a way that for the first 20 metres he goes with a speed of 20 m/s and the next 20 metres with a speed of 10 m/s. Similarly he continuous for the rest of the hexagonal path with the same alternating speeds *i.e.*, 20 m/s and 10 m/s. The average speed of the artist per round of the circus is :
(a) 13.33 m/s (b) 23.66 m/s
(c) $39\frac{1}{3}$ m/s (d) 15 m/s

15 The average of 26, 29, n, 35 and 43 lies between 25 and 35. If n is always an integer and greater than the average of the given integers then the value of n is :
(a) $33 < n < 47$ (b) $34 < n < 43$
(c) $33 < n < 42$ (d) none of these

16 The average of 4 distinct prime numbers a, b, c and d is 35, where $a < b < c < d$. a and d are equidistant from 36 and b and c are equidistant from 34 and a, b are equidistant from 30 and c and d are equidistant from 40. The difference between a and d is :
(a) 30 (b) 14
(c) 21 (d) can't be determined

17 There are four numbers whose product is 9261000 and each of these four numbers is formed by 3 distinct prime numbers. The average of all the four numbers is :
(a) 61.75 (b) 67.25
(c) 82.33 (d) data insufficient

18 A number is such that when it is multiplied by 8, it gives another number which is as much above from 270 as the original number (itself) is below 270. The average of the original number and the resultant number is :
(a) 33.75 (b) 190
(c) can't be determined (d) none of these

19 In the above question (no. 18) the average of the original no. and 270 is :
(a) 165 (b) 185
(c) 135 (d) none of these

20 In a particular week the average number of people who visited the Tajmahal is 40. If we exclude the holidays then the average is increased by 16. Further if we exclude also the day on which the maximum number of 112 people visited the Tajmahal, then the average becomes 42. The number of holidays in the week is :
(a) 1 (b) 2
(c) 3 (d) data insufficient

21 The average age of all the 20 students of a class is 24. The minimum age of a student is 18 and the maximum age of another student in the same class is 30 years. When the two students whose average age was 26 years resticated from the class but later on one of the resticated student was readmitted. Now the average age of the class is :
(a) 23.89 years (b) 28.39 years
(c) 25 years (d) can't be determined

22 In a set of prime and composite numbers, the composite numbers are twice the number of prime numbers and the average of all the numbers of the set is 9. If the number of prime numbers and composite numbers are exchanged then the average of the set of numbers is increased by 2. If during the exchange of the numbers the average of the prime numbers and composite numbers individually remained constant, then the ratio of the average of composite numbers to the average of prime numbers (initially) was :
(a) $\frac{7}{13}$ (b) $\frac{13}{7}$
(c) 9/11 (d) none of these

23 The total age of all the guests in the party was 540 years. If a South Indian couple (guests) left the party, then the average of the remaining guests still remained unchanged, where the age of both the husband and wife (the South Indian couple) was same, then the average age of this couple and the total number of guests in the party, respectively, can be :
(a) 18, 27 (b) 20, 27
(c) 15, 38 (d) can't be determined

24 In the command hospital Lucknow the sum of the ages of all the 29 people *i. e.*, physicians, surgeons and nurses is 696. If the age of each physician, each surgeon and each nurse be 1 year, 6 years and 3 years more, than the average age of the whole staff would have been 3 years more. If the number of surgeon is a square root of a two digit number which is also a perfect cube, then the number of nurses in the hospital is :
(a) 12 (b) 15
(c) 16 (d) none of these

25 The average expenditure of Sarvesh for the January to June is ₹ 4200 and he spents ₹ 1200 in January and ₹ 1500 in July. The average expenditure for the months of February to July is :
(a) 4250 (b) 4520
(c) 4060 (d) none of these

26 The average marks of Sameer decreased by 1, when he replaced the subject in which he has scored 40 marks by the other two subjects in which he has just scored 23 and 25 marks respectively. later he has also included 57 marks of Computer Science, then the average marks increased by 2. How many subjects were there initially?
(a) 6 (b) 12
(c) 15 (d) can't be determined

27 In a combined family the average age of 4 males and 7 females is 42 and 20 years respectively. If two persons whose average age is 13 years have left the family and other three people joined the family whose respective ages are 11, 15 and 28 years, then the average age of the new family is increased by :
(a) 4 years (b) 1 year
(c) 3 years (d) none of these

28 A teacher gave sum to his class to find the average of n numbers viz. 1, 2, 3, 4, 5, 6 ... etc. But when the teacher checked the solution, he has found that during the calculation a student just missed a number for the addition thus his average of the n numbers was 15. The value of n is :
(a) 30 (b) 26
(c) 31 (d) not unique

29 The average earning of a group of persons is ₹ 50 per day. The difference between the highest earning and lowest earning of any two persons of the group is ₹ 45. If these two people are excluded the average earning of the group decreases by ₹ 1. If the minimum earning of the person in the group lies between 42 and 47 and the number of persons initially in the group was equal to a prime number, with both its digits prime. The number of persons in the group initially was :
(a) 29 (b) 53
(c) 31 (d) none of these

30 There are three categories of jobs A, B and C. The average salary of the students who got the job of A and B categories is 26 lakh per annum. The average salary of the students who got the job of B and C category is 44 lakh per annum and the average salary of those students who got the job of A and C categories is 34 lakh per annum. The most appropriate (or closest) range of average salary of all the three categories (if it is known that each student gets only one category of jobs *i. e.*, A, B and C) :
(a) lies between 30 and 44
(b) lies between 28 and 34
(c) lies between 34 and 43
(d) lies between 29 and 48

31 Out of the five integral numbers C is the average of A and D. B is greater than C and less than D. Also, B is the average of A and E. The middle most number in the sequence is :

(a) A (b) B
(c) C (d) D

32 The average age of Donald, his wife and their two children is 23 years. His wife is just 4 year younger than Donald himself and his wife was 24 years old when his daughter was born. He was 32 years old when his son was born. The average age of Donald and his daughter is :

(a) 25 years
(b) 22.5 years
(c) 26 years
(d) can't be determined

33 There are only five people in the Aman Verma's family. Aman, his wife, a son and two daughters. The younger daughter's age is $\frac{4}{5}$th of the elder daughter's age. The age of eldest daughter is $\frac{3}{8}$ times that of her father Aman and the age of the son is $\frac{1}{5}$th that of his father Aman. 4 years ago the age of her wife was 8 times that of his son and now the sum of the ages of the younger daughter and wife is same as the sum of the ages of Aman and his son. The average age of the family is :

(a) 22.22 years (b) 25.4 years
(c) 21.2 years (d) none of these

34 The average weight of a political party is decreased by 1, when some new politicians joined the party, whose strength is $\frac{1}{4}$th of the existing (or old) politicians and the total weight of the new politicians is 209 kgs. What is the new average weight of all the politicians if it is known that in any case the number of politicians always must be greater than 50 but less than 100 :

(a) 15 kgs (b) 16 kgs
(c) 18 kgs (d) 19 kgs

35 Ravi went to Kanpur from Lucknow by his four wheeler. During the journey he had to use the spare wheel (*i. e.*, stepney). Thus he finished his 160 km journey. The average distance covered by the wheels of his car is :

(a) 40 km (b) 120 km
(c) 128 km (d) 48 km

36 There are 6 consecutive odd numbers in increasing order. The difference between the average of the squares of the first 4 numbers and the last four numbers is 64. If the sum of the squares of the first and the last element (*i. e.*, odd numbers) is 178, then the average of all the six numbers is :

(a) 7 (b) 8
(c) 9 (d) 10

37 The average age of board of directors of a company, having 10 directors was 48 years. Coincidentally when a director aged 53 resigned from the board of directors, another director died on the same day. So a new director joined the board of directors aged 34. Next year in the same month the average age of all the 9 directors was found to be 46 years. The age of the late (*i. e.*, dead) director at the time of his death was :

(a) 56 years (b) 53 years
(c) 57 years (d) 61 years

38 In an office the average weight of 24 employees is 60 kg. If n employees were included whose average weight was 54 kg, then the total number of employees in the office, (given that the new average weight of all the $(24 + n)$ employees is a whole number) :

(a) 36 (b) 30
(c) 34 (d) 25

39 The average age of 100 nurses in a nursing home in 1982 was 50 years. In 1984, 20 nurses retired from their job, whose average age was 60 years. After a huge gap in 1987, 40 nurses were employed whose average age was 38 years. The average age of all the nurses in 1990 was :

(a) 53 years
(b) 51 years
(c) 48.5 years
(d) data insufficent

Directions (for Q. Nos. 40 to 44) *Bhartiya Idol is a talent search programme launched and run by TV Tarana. In this programme each participating candidate has to appear for the audition in such a manner that a candidate plays the Antakchhary with its aspiring opponent until he fails, otherwise he can continue and become winner. If a candidate fails on his part, he has to leave the contest and in place of it another candidate starts off with the same existing opponent. The number of points a candidate scores is equal to the number of times he responds correctly to his opponents. The candidate who finishes the game (or who wins over the last opponent) is declared as a winner, even if he/she scores less points and starts as the last participant. The points scored individually by all the 10 candidates are as shown below. Besides, if a person loses the contest only if he/she responds incorrectly.*

	Participants	Points
1.	Rajesh	8
2.	Radhe	10
3.	Harish	7
4.	Prajakta	12
5.	Aditya	25
6.	Rahul	24
7.	Shekhar	18
8.	Amit Tandon	23
9.	Amit Sana	29
10.	Abhijeet	30

40 The average score of all the 10 participants is :
(a) 21.5 (b) 18.6
(c) 18.3 (d) none of these

41 A candidate can face maximum 'n' opponents. The value of 'n' is :
(a) 3 (b) 4
(c) 1 (d) can't be determined

42 The average score of five participants who started earlier :
(a) 23 (b) 12
(c) 19.5 (d) can't be determined

43 If Abhijeet has not started off earlier and he is the winner then the minimum number of participants who has lost the game before he has started off :
(a) 4 (b) 6
(c) 5 (d) none of these

44 For the above question the maximum possible average of the opponents of Abhijeet could be :
(a) 13.5 (b) 9.25 (c) 13.0 (d) 17.5

45 In an NGO, the daily average wages of 20 illiterate employees is decreased from ₹ 25 to ₹ 10, thus the average salary of all the literate (educated) and illiterate employees is decreased by ₹ 10 per day. The number of educated employees working in the NGO is :
(a) 15 (b) 20
(c) 10 (d) data insufficient

46 Mr. Tyagi while going from Meerut to Saharanpur covered half the distance by train at the speed of 96 km/hr then he covered half of the rest distance by his scooter at the speed of 60 km/hr and finally he covered the rest distance at the speed of 40 km/hr by car. The average speed at which Mr. Tyagi completed his journey is :
(a) 64 km/hr (b) 56 km/hr
(c) 60 km/hr (d) 36 km/hr

47 There are four types of candidates in our coaching preparing for the CAT. The number of students of Engineering, Science, Commerce and Humanities is 40, 60, 50 and 30 respectively and the respective percentage of students who qualified the CAT is 80%, 75%, 60% and 50% respectively the overall percentage of successful candidates in our institute is :
(a) 67.77% (b) 66.66%
(c) 68.5% (d) none of these

48 Mr. Manmohan calculated the average of 10, 'three digit numbers'. But due to mistake he reversed the digits of a number and thus his average increased by 19.8. The difference between the unit digit and hundreds digit of that number is :
(a) 8
(b) 4
(c) 2
(d) can't be determined

49 Once my peon went to the office of SKYLINE COURIER with 4 different envelopes. The clerk in the office measured the weights in all possible pairs. The weights obtained are 59 gm, 61 gm, 62 gm, 63 gm, 64 gm and 66 gm. The weight of the heaviest envelope is :
(a) 35 gm (b) 36 gm
(c) 34 gm
(d) can't be determined

50 The average expenditure of the hotel when there are 10 guests is ₹ 60 per guests and the average expenditure is ₹ 40 when there are 20 guests. If it is known that there are some fixed expenses irrespective of the number of guests then the average expenditure per guest when there are 40 guests in the hotel :
(a) ₹ 30 (b) ₹ 25
(c) ₹ 20 (d) can't be determined

LEVEL 03 > Final Round

Directions (for Q. Nos. 1 to 15) *There are 3 sets of natural numbers 1 to 100. Set A contains all the natural numbers which are prime, upto 100. Set B contains all the non-prime even natural numbers upto 100. Set C contains all the non-prime odd natural numbers upto 100 i.e.,*

$A = \{2, 3, 5, 7, 11, \ldots 89, 97\}$
$B = \{4, 6, 8, 10, 12, \ldots 98, 100\}$
$C = \{1, 9, 15, 21, 25, 27, 33, \ldots 95, 99\}$

1 The average of all the elements of A, B and C is :
(a) 49.50 (b) 50.50
(c) 55 (d) none of these

2 The average of all the elements of B is :
(a) 52 (b) 48
(c) 49 (d) none of these

3 If the average (factually correct) of the set A is 42.46, then the average of the Set C is :
(a) 52
(b) 49.87
(c) 55.46
(d) can't be determined

4 The average of the elements of the Set A and C combined is:
(a) 49.0588 (b) 49.0372
(c) 50 (d) none of these

5 If an element less than 50 belongs to Set A is transferred to Set B, then the average of Set B :
(a) increases (b) decreases
(c) remains constant (d) can't be determined

Answers

Level 01 *Basic Level Exercise*

1 *(c)*	2 *(b)*	3 *(a)*	4 *(b)*	5 *(b)*	6 *(a)*	7 *(a)*	8 *(c)*	9 *(c)*	10 *(b)*
11 *(a)*	12 *(a)*	13 *(b)*	14 *(a)*	15 *(c)*	16 *(c)*	17 *(c)*	18 *(a)*	19 *(b)*	20 *(b)*
21 *(d)*	22 *(c)*	23 *(d)*	24 *(b)*	25 *(c)*	26 *(a)*	27 *(b)*	28 *(b)*	29 *(a)*	30 *(b)*
31 *(c)*	32 *(a)*	33 *(b)*	34 *(b)*	35 *(b)*	36 *(a)*	37 *(a)*	38 *(b)*	39 *(d)*	40 *(a)*
41 *(a)*	42 *(c)*	43 *(b)*	44 *(a)*	45 *(d)*	46 *(a)*	47 *(a)*	48 *(a)*	49 *(b)*	50 *(c)*
51 *(b)*	52 *(b)*	53 *(d)*	54 *(d)*	55 *(d)*	56 *(a)*	57 *(c)*	58 *(a)*	59 *(b)*	60 *(a)*

Level 02 *Higher Level Exercise*

1 *(a)*	2 *(c)*	3 *(b)*	4 *(b)*	5 *(b)*	6 *(c)*	7 *(d)*	8 *(b)*	9 *(d)*	10 *(a)*
11 *(c)*	12 *(d)*	13 *(d)*	14 *(a)*	15 *(c)*	16 *(b)*	17 *(a)*	18 *(d)*	19 *(a)*	20 *(b)*
21 *(d)*	22 *(a)*	23 *(b)*	24 *(d)*	25 *(a)*	26 *(c)*	27 *(d)*	28 *(d)*	29 *(d)*	30 *(a)*
31 *(b)*	32 *(a)*	33 *(a)*	34 *(a)*	35 *(c)*	36 *(b)*	37 *(a)*	38 *(a)*	39 *(b)*	40 *(b)*
41 *(b)*	42 *(d)*	43 *(c)*	44 *(a)*	45 *(c)*	46 *(a)*	47 *(a)*	48 *(c)*	49 *(c)*	50 *(a)*

Level 03 *Final Round*

1 *(b)*	2 *(a)*	3 *(c)*	4 *(a)*	5 *(b)*	6 *(c)*	7 *(d)*	8 *(a)*	9 *(c)*	10 *(c)*
11 *(b)*	12 *(b)*	13 *(a)*	14 *(c)*	15 *(d)*	16 *(a)*	17 *(d)*	18 *(b)*	19 *(a)*	20 *(c)*
21 *(c)*	22 *(a)*	23 *(b)*	24 *(c)*	25 *(c)*	26 *(a)*	27 *(d)*	28 *(a)*	29 *(b)*	30 *(c)*
31 *(d)*									

Hints & Solutions

Level 01 *Basic Level Exercise*

1 Initially the total weight = $20 \times 45 = 900$

Now, the total weight when a student has been replaced

$$= 20 \times 44 = 880$$

It means the weight of the new student is 20 kgs less than the replaced student. Hence the weight of the replaced student = 40 + 20 = 60 kgs

Alternatively Since you know that there are total 20 students and when their average weight decreases by 1, it means on an average 1 kg weight gets reduced from each of the students. Thus there is 20 kg weight loss, in total. Again this happens due to the student whose weight is 40 kg who replaces an old student. Thus we can say that the weight of the old (or existing) student of the class was 60 kg, which is reduced by a 40 kg student.

2 The only useful data is that the average length of the thumb and index finger is 2.8 and the length of index finger is 3 inches. Now the total length of index finger and thumb

$$= 2.8 \times 2 = 5.6 \text{ inches}$$

Thus the length of thumb is $(5.6 - 3) = 2.6$ inches.

3 The average price of the new mixture

$$= \frac{(38 \times 8) + (43 \times 4) + (49 \times 3)}{15} = 41.53$$

4 Average speed when Pankaj was returning

$$= \frac{2 \times 10 \times 30}{40} = 15 \text{ km/hr}$$

Now the average speed of the whole journey

$$= \frac{2 \times 15 \times 60}{75} = 24 \text{ km/hr}$$

5 Use unitary method :

In $\frac{1}{15}$ hours, 123 typists can type 984 papers

In 1 minute, 123 typists can type $\frac{984}{4} = 246$ papers

In 1 minute, 1 typist can type $\frac{246}{123} = 2$ papers

Hint $\frac{1}{15}$ hours = 4 minutes

6 Average cost $= \frac{20 \times 3x + 15 \times 2x + 18 \times 4x}{9x} = 18$

7 Earning in the 8 months = $600 \times 8 = 4800$

Earning in the 4 months $= \left(600 \times \frac{3}{2}\right) \times 4 = 3600$

Total earning = ₹ 8400

Saving in 8 months = $400 \times 8 = 3200$

Saving in 4 months $= \left(400 \times \frac{5}{4}\right) \times 4 = 2000$

Total savings = 5200

Total expenditure for 12 months

$$= 8400 - 5200 = 3200$$

Therefore, average saving per month $= \frac{3200}{12} = 266.66$

8 There are 6 prime numbers between 47 and 74 as given below :

53, 59, 61, 67, 71, 73

There are only two combinations whose average is $\frac{191}{3}$ or whose total sum is 191.

These are, {53, 67, 71} and {59, 61, 71}

We observe that in the first set the difference between greatest and smallest prime number is 18.

Hence, choice (c) is correct.

9 Go back to the fundamentals and see the property number 4 and 2 of averages.

Hence, required average = $11 \times 5 + 5 = 60$

10 The average score after 48th innings = 48

The average score after 49th innings = 49

Since, $\frac{(48 \times 48 + 97)}{49} = 49$

Now the requirement of runs = $49 + (50 \times 2) = 149$

or $(50 \times 51) - (49 \times 49) = 149$

11

Average			Total
$S + G$	$\rightarrow$	35	70
$K + G$	$\rightarrow$	32	64
$S + K$	$\rightarrow$	38	76
$S + K + G$	$\rightarrow$	35	105
$D + I$	$\rightarrow$	$\frac{35}{2}$	35

$\therefore \quad \frac{S + K + G + D + I}{5} = \frac{105 + 35}{5} = 28$

12 Kaif $\rightarrow$ 35

Sachin $\rightarrow$ 41

Ganguli $\rightarrow$ 29

Dhoni < 35 and Irfan < 35

Hence, Sachin is the senior person.

13 It is the same as a person with 20 years of more age replaces an existing person of the group (or village)

Since the total age of the village having n persons, is being increased by 20 years and the average age of village is being increased by 1 year, hence there are total 20 people in the village.

Alternatively $(n \times 42) + 20 = (n \times 43)$

$n = 20$

14 Average rainfall $= \dfrac{2\times 6 + 4\times 5 + 4\times 10 + 2\times 6}{12} = 7$

15 $400 \times 7 = (300 \times 3) + (250 \times 2) + (n \times 2) \Rightarrow 700$

16 The new player must be 11 kg heavier than the replaced one. Hence the weight of the new person = 55 + 11 = 66 kg.

17

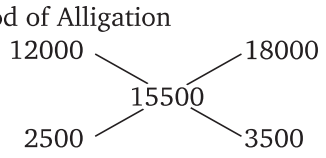
	No. of family members	Average age	Total age
4 years ago	6	25	150
Presently	6	29	174

But the no. of family members (presently) = 7
and average age (presently) = 25
Therefore the total age = $25 \times 7 = 175$
Hence, the age of child = 175 – 174 = 1 year

18 Amitabh's total expenditure for Jan-June
$= 4200 \times 6 = 25200$
Expenditure for February-June = 25200 – 1200 = 24000
Expenditure for the months of February-July
= 24000 + 1500 = 25500
The average expenditure $= \dfrac{25500}{6} = 4250$

19. Total value of $a + 11 + 23 + 17 = 60 \Rightarrow a = 9$
Again $a + b + 12 + 25 = 64 \Rightarrow b = 18$
Therefore, $\dfrac{a}{b} = \dfrac{1}{2}$

20 By the method of Alligation



Therefore the ratio of number of Junior level employee to the senior level employee = 5 : 7
Hence the required fraction $= \dfrac{5}{(5+7)} = \dfrac{5}{12}$

Alternatively Go through options

$$\frac{J}{J+S} = \frac{5x}{12x} \Rightarrow \frac{J}{S} = \frac{5x}{7x}$$

Therefore $15500 \times 12x = 12000 \times 5x + 18000 \times 7x$
LHS = RHS, Hence correct

21 None of *a, b, c* is correct.
The correct answer can be $\dfrac{a+b+c+d+e}{5}$
or $\dfrac{a+e}{2} = \dfrac{b+d}{2} = c$
Where *a, b, c, d* and *e* are consecutive odd numbers.

22 $31 \times 31 - 30 \times 30 = 61$ years

23 $a + b + c + d + e = 5 \times 92 = 460$
$a + b + c = 3 \times 83 = 249$
$c + d + e = 3 \times 97 = 291$
$\therefore \quad c = (a + b + c) + (c + d + e) - (a + b + c + d + e)$
or $c = 540 - 460$ or $c = 80$

24 $19 \times 26 - 18 \times 25 = 44$ years

25 $\dfrac{k + k + 2 + k + 4 + k + 6 + k + 8 + k + 10 + k + 12}{7}$
$= k + 6$

26 $\dfrac{\frac{x}{4}\times\frac{85}{100} + \frac{x}{3}\times\frac{70}{100} + \frac{5x}{12}\times\frac{56}{100}}{x} \times 100 = 67.916$
Here, x is the total number of students in the whole class.

27 Since there are five possibilities or combinations (as ${}^5C_4 = 5$)
Therefore, average $= \dfrac{5\times 600}{4} = 750$ mm

28 $\dfrac{\frac{23}{5} + \frac{8}{3} + \frac{62}{9} + \frac{112}{15} + \frac{32}{9}}{5} = 5\dfrac{8}{225}$

29 $\dfrac{1000.0001 + 100.001 + 10.01 + 1.1}{4}$
$= \dfrac{1111.1111}{4} = 277.777$

30 $\dfrac{7 + 8 + 9 + 10 + 11 + 12 + 13}{7} = 10 \Rightarrow \dfrac{7 + 13}{2} = 10$

31 $\left(\dfrac{1 + 2 + 3 + \ldots + 100}{100}\right) = \dfrac{\left(\frac{100\times 101}{2}\right)}{100} = 50.50$

32 $\left(\dfrac{1 + 3 + 5 + 7 + \ldots + 99}{50}\right) = \left(\dfrac{50 \times 50}{50}\right) = 50$

33 $\dfrac{2 + 4 + 6 + \ldots + 198}{99} = \dfrac{99 \times 100}{99} = 100$

34 $a + b + c = 237$
$a + c = 158$
$\Rightarrow \quad b = 79$

35 $\dfrac{M + A + S}{3} = 11111$
Also, $\dfrac{S + M}{2} = 11111 \quad (\because M - 11111 = 11111 - S)$
$\Rightarrow \quad A = 11111$

36 $\dfrac{2\times 200 \times 600}{800} = 300$ km/hr

37 $\dfrac{A + B + C + D}{4} = \dfrac{400 \times 2 + 600 \times 2}{4} = 500$

38 Average weight $= \dfrac{20 + 50}{100} = 0.7$ kg ($\because$ 1 kg = 1000 gm)

39 Average $= \dfrac{770}{11} = 70$

40 $\dfrac{3x + 4x + 5x}{3} = 12 \quad \Rightarrow \quad x = 3$
$\therefore$ The ages are 9, 12 and 15 years
and the required average $= \dfrac{9 + 15}{2} = 12$

41 Since the salary of each employee is being increased therefore the average salary will also increased by ₹ 2000. Thus the required average = 22000.

42 $11 \times 21 - 10 \times 20 = 31$

43 $20 \times 180 - 19 \times 178 = 218$

44 Let the price of $A > B > C$

Then, $\dfrac{A+B}{2} = 2C$

$A + B = 4C$

Now, $A + B + C = 5 \times 3 = 15$ crore

$\Rightarrow \quad 5C = 15$ crore

$\therefore \quad C = 3$ crore

45 Since, we don't know about the price of B.

46 $\left(\dfrac{40+70+142}{50+100+150}\right) \times 100 = 84\%$

47 Since 1 is neither prime nor composite number. Thus there are only 99 number viz. 2, 3, 4, 5, 6, …, 99, 100.

Hence $= \left(\dfrac{2+3+4+5+6+\ldots+100}{99}\right)$

$= \left(\dfrac{(1+2+3+4+\ldots+100)-1}{99}\right)$

$= \dfrac{5050-1}{99} = \dfrac{5049}{99} = 51$

48 $\dfrac{1+4+9+16+\ldots+100}{10} = \left(\dfrac{10 \times 11 \times 21}{6 \times 10}\right)$

$\left[\because 1^2 + 2^2 + \ldots + n^2 = \dfrac{n(n+1)(2n+1)}{6}\right] = 38.5$

49 The non-negative integers upto 99 are 0, 1, 2, 3, …, 99.

Therefore, average $= \left(\dfrac{0+1+2+3+\ldots+99}{100}\right)$

$= \dfrac{\left(99 \times \dfrac{100}{2}\right)}{100} \qquad \left[\because 1+2+3+\ldots+n = \dfrac{n(n+1)}{2}\right]$

$= 49.5$

50 $\dfrac{7+14+21+\ldots 77}{11} = \dfrac{7(1+2+\ldots+11)}{11}$

$= \dfrac{7 \times 11 \times 12}{11 \times 2} = 42 \qquad \left[\because 1+2+3+\ldots+n = \dfrac{n(n+1)}{2}\right]$

Alternatively Since, all the numbers are in A.P. Further there are odd number of numbers (*i. e.*, 11) in the sequence. Thus the middle most term is the average of the sequence, which is 42.

51

	Average	Total
A, B, C, D	40	160
A, B, C, D, E	41	205
F, B, C, D, E	42	210

Thus, $E = 205 - 160 = 45$

Hence, $F, B, C, D = 210 - 45 = 165$

Therefore, average of $F, B, C, D = 41.25$

52 Three consecutive natural numbers whose average is k are $(k-1), (k), (k+1)$.

The next two numbers will be $(k+2)$ and $(k+3)$.

Therefore, average of $(k-1), (k), (k+1)(k+2)$ and $(k+3)$ is $(k+1)$.

Alternatively Consider any 3 consecutive natural number *eg.* 4, 5, 6 the average is 5.

Again 4, 5, 6, 7, 8 the average is 6.

To verify it consider some different numbers *eg.* 10, 11, 12 average = 11 and 10, 11, 12, 13, 14 average = 12

Hence, proved that the average is increased by 1.

53 The 5 consecutive odd numbers whose average is k are $(k-4), (k-2), k, (k+2), (k+4)$

Again the average of $(k-4), (k-2), (k), (k+2), (k+4), (k+6), (k+8)$ is $(k+2)$

Alternatively Consider some appropriate numbers.

54 The increase in weight $= (5 \times 42.5) - (5 \times 42) = 2.5$ kg

But we don't know the weight of the replaced officer. So, we can't determine.

55 The decrease in age $= 6 \times (28 - 27) = 6$ years

But we don't know the age of the old servant which is being replaced. So, we cannot determine the average age of new servant.

56 The age of new servant $= 31 - 6 = 25$ years

57 $A + B + C = 12000 \times 3$

$B + C + D = 15000 \times 3$

$\Rightarrow \quad D - A = 3000 \times 3$

$D - A = 9000$

also $D = 2A$

$\Rightarrow \quad D = 18000$ and $A = 9000$

Therefore, average salary of B and $C = \dfrac{(45000 - 18000)}{2}$

$= 13500$

58 The price of the costliest and cheapest computer

$= (80 \times 30000) - (78 \times 29500) = 99000$

Therefore the price of the cheapest computer

$= 99000 - 80000 = 19000$

59 $\dfrac{10 \times 50 + 50 \times 10 + 20 \times 25 + 25 \times 20}{4} = 500$

60

	Auto Rickshaw	Maxi. cab	Mini. Bus
No. of seats	4	10	20
No. of seats occupied	4	8	15
Rate per seat	12	15	8
Total amount (in ₹)	48	120	120

Therefore, average earning

$= \dfrac{48 + 120 + 120}{3} = 96$

☑ Level 02 *Higher Level Exercise*

1 For this type of questions take the LCM of speeds and assume the LCM as the distance.

Then the time taken @ speed of 60 km/hr $= \frac{300}{60} = 5$ hrs

Again the time taken @ speed of 50 km/hr $= \frac{300}{50} = 6$ hrs

Thus we see that in place of 5 hrs train takes 6 hours. It means the train takes 1 hour extra and this one hour is stopping period in the total time of 6 hours. Thus in 6 hour train halts for 1 hour. So in 1 hour train will stop for $\frac{1}{6}$ hours or 10 minutes.

Alternatively (short cut)

Halting (or stopping) time $= \left(1 - \frac{\text{slower speed}}{\text{faster speed}}\right)$ hours

$= 1 - \frac{50}{60} = \frac{10}{60} = \frac{1}{6}$ hours

$= 10$ minutes

($\because$ 1 hour = 60 minutes)

2 $$\frac{[r + (p - q)] + p + q}{3} = 21 + \frac{p + q + r}{3}$$

$$\Rightarrow \quad \frac{2p + r}{3} - 21 = \frac{p + q + r}{3}$$

$$\Rightarrow \quad \frac{p - q}{3} = 21$$

$$\Rightarrow \quad p - q = 63$$

3 Let the speed for the first hour be x km/hr

Then the speed for the second hour be $\frac{7}{5}x$ km/hr

Then the speed for the third hour be $\frac{10}{7} \times \frac{7}{5}x = 2x$ km/hr

Then the speed for the fourth hour be

$$2x \times \frac{7}{5} = \frac{14x}{5} \text{ km/hr}$$

Therefore, total distance in four hours

$$= x + \frac{7}{5}x + 2x + \frac{14x}{5} = \frac{36x}{5} \text{ km}$$

$$\therefore \text{ Average speed} = \frac{\text{Total Distance}}{\text{Total Time}} = \frac{\left(\frac{36x}{5}\right)}{4}$$

$$= \frac{9x}{5} \text{ km/hr}$$

Again the distance in 4 hours @ speed of x km/hr.

Which is half of the third hour's speed is $4x$ km.

Hence, $$\frac{36x}{5} - 4x = 160 \text{ km}$$

$$\Rightarrow \quad x = 50$$

Hence, the average speed $= \frac{9 \times 50}{5} = 90$ km/hr

4 The total score of 3 toppers

$$= 123 \times 120 - 120 \times 118.5$$

$$= 540$$

The highest possible score of the third highest topper is possible when the score of other two toppers was minimum

So,

$$\left.\begin{array}{l} 1^{st} \text{ rankers score} = 187 \text{ (minimum)} \\ 2^{nd} \text{ rankers score} = 186 \text{ (minimum)} \\ 3^{rd} \text{ rankers score} = 167 \text{ (maximum)} \end{array}\right\} 540$$

5 Go through options : $40 \times 35 + 60 \times 25 = 29 \times 100$

Since there are 40 ladies and 60 gents

Alternatively $40 \times 7x + 60 \times 5x = 29 \times 100$

$$\Rightarrow \quad x = 5 \quad \therefore \quad 7x = 35$$

6

	Lok Sabha	Rajya Sabha
No. of MPs	300	200
Ave. Age	40	50

Since, when a member of Rajya Sabha joins the Lok Sabha and the average age of both the houses increases, it means the average age of this member must lie between 40 and 50. When the age of this member is greater than 40, then the average age of the Lok Sabha increases. Again when the age of this member is less than 50, then after leaving it, the average age of the Rajya Sabha increases.

Solutions (for Q. Nos. 7 to 9)

	No. of family members	Average	Total
8 years ago →	5	36	180
Presently →	(if) 5	(36 + 8) = 44	220
	7	36	252

7 From the above explanation we have no any clue about his wife's age.

8 Since we know that the difference between the age of any two persons remains always constant, while the ratio of their ages gets changed as the time changes.

So, if the age of his child be x (presently)

Then the age of wife be $x + 26$ (presently)

Thus, the total age $= x + (x + 26) = 32$ [$\because 252 - 220 = 32$]

$$\Rightarrow \quad x = 3$$

Therefore the age of her child is 3 years and her self is 29 years. Hence her age at the time of the birth of her child was 26 years.

Alternatively As we have mentioned above that the age difference remains always constant. Therefore her age at the time of her child's birth was 26 years.

9 Since there is no clue. So, we can't determine.

10

	No. of family members	Average	Total
Eleven years earlier	4	28	112
Presently	if 4	39	156
	6	28	168

Since it is obvious that just after the birth of the youngest member (*i. e.*, child) was 6 family members in the family. Therefore at the time of the birth of the youngest child the elder child's age was 6 years.

Now the sum of their ages $= x + (x + 6) = 12$

$= (168 - 156)$

$\Rightarrow \quad x = 3 \quad \text{and} \quad (x + 3) = 9$

11 Go through options

$\frac{6}{4} = 1.5$ hour, when he was walking

Therefore, time of journey by train $= 3 - 1.5 = 1.5$

Now, the distance travelled by train

$= 1.5 \times 60 = 96 - 6 = 90$

Hence correct.

Alternatively Total distance $= 32 \times 3 = 6 + 60 \times x$

$\Rightarrow \quad x = 1.5$ hours

Thus, the speed of walking $= \frac{6}{1.5} = 4$ km/hr

12 Let the number of total MPs $= n$ and their average age be x then

$$n \times x = \frac{4}{5} n \times \frac{5}{4} x + \frac{n}{5} \times y$$

$\Rightarrow \quad$ either $n = 0$ or $y = 0$

[Since, there are only 80% MPs remained in the house which is equal to $\frac{4}{5} n$ and the increase in average age $= 20\% = \frac{5}{4} x$]

Thus, there cannot be any possible value of n.

13 $nx = \frac{4}{5} n \times \frac{5}{4} x + \frac{n}{5} \times 30$

$\Rightarrow \quad n = 0$ which is impossible

So, there is no any woman MP in the Lok Sabha.

14 Average speed $= \frac{\text{Total distance}}{\text{Total time}}$

$= \frac{6 \times 20}{9} = 13.33$ m/s



Hexagonal path has six sides

15 Average of 26, 29, 35 and 43 is 33.25. Also the average of 26, 29, n, 35 and 43 lies between 25 and 35 *i. e.*,

$$25 < \frac{26 + 29 + n + 35 + 43}{5} < 35$$

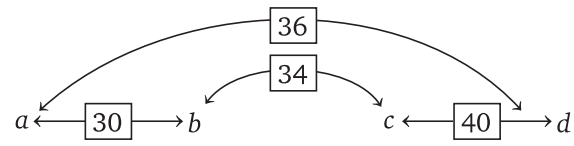
$\Rightarrow \quad 125 < 26 + 29 + n + 35 + 43 < 175$

$\Rightarrow \quad 125 < 133 + n < 175$

$\Rightarrow \quad n < 42$

Since the value of n is an integer and greater than 33.25, then $33 < n < 42$; for every integer n.

16 Given that $a < b < c < d$



The only possible prime number between 30 and 34 is 31

Hence, $b = 31$, therefore $a = 29$

Similarly, $c = 37$ and $d = 43$

Therefore, $d - a = 43 - 29 = 14$

17 $9261000 = 2^3 \times 3^3 \times 5^3 \times 7^3$

$= (2 \times 3 \times 5) \times (2 \times 3 \times 7) \times (2 \times 5 \times 7) \times (3 \times 5 \times 7)$

$= 30 \times 42 \times 70 \times 105$

Therefore the average of 30, 42, 70 and 105 is 61.75.

18 Since the number is as below 270 as its multiple is as above 270. It means these two numbers are equidistant from 270. Hence their average is 270.

Alternatively Let the number be x then.

$270 - x = 8x - 270$

$\Rightarrow \quad x = 60 \quad \text{and} \quad 8x = 480$

Therefore the average of 60 and 480 is 270.

19 The average of 60 and 270 is 165

20 Number of days in a week = 7

Average number of visitors = 40

Total visitors = 280 (= 7 × 40)

Now, if n be the number of holidays in a week, then

$(7 - n) \times 56 = 280 \qquad (40 + 16 = 56)$

$\Rightarrow \quad n = 2$

The rest data is redundant or useless, since our problem is solved here without using it.

Alternatively Go through options

$40 \times 7 = 280 = 56 \times 5$ hence proved.

21 Since we don't know their ages individually, so we cannot calculate the average of the class when a student of unknown age readmitted in the class.

22 Let the average of prime numbers be P and average of composite numbers be C.

Again the number of prime numbers be x, then the number of composite numbers be $2x$.

Then, $\frac{Px + 2Cx}{3x} = 9 \quad \Rightarrow \quad P + 2C = 27 \qquad \ldots(i)$

and $\dfrac{2Px + Cx}{3x} = 11$

$\Rightarrow$ $2P + C = 33$ …(ii)

On adding eqs. (i) and (ii) we get $P + C = 20$

and on subtracting eqs. (i) from (ii), we get $P - C = 6$

Therefore, $P = 13$ and $C = 7$

Thus, $\dfrac{C}{P} = \dfrac{7}{13}$

23 Solve through option $\dfrac{540}{27} = 20$

$$\frac{540 - (2 \times 20)}{25} = \frac{500}{25} = 20$$

Hence option (b) is correct.

24 Number of members in the staff × average age = Total age

$$29 \times 24 = 696 \Rightarrow 29 \times 27 = 783$$

Hence, change in total age = 87

	Physician	Surgeon	Nurse
No. of members	$(21 - x)$	8	x
Increase in average age	1	6	3
Increase in total age	$(21 - x)$	48	$3x$

Since $P + S + N = 29$

Therefore, if there would be x nurses, then there must be $(21 - x)$ physicians. Again, total change (or increase) in age

$$= (21 - x) + 48 + 3x = 87 \Rightarrow x = 9$$

25 Total exp. Jan-June = $4200 \times 6 = ₹\ 25200$

Total exp. Feb-June = $25200 - 1200 = ₹\ 24000$

Total exp. Feb-July = $24000 + 1500 = ₹\ 25500$

The average expenditure Feb-July = $\dfrac{25500}{6} = 4250$

26 Let the number of subjects be n and average marks be x, then total marks = nx

Again $(n + 1)(x - 1) = (nx - 40) + (23 + 25)$

$\Rightarrow$ $x - n = 9$ …(i)

Further $(n + 2)(x + 1) = (nx - 40) + (23 + 25) + 57$

$\Rightarrow$ $nx + 2x + n + 2 = nx + 65$

$\Rightarrow$ $2x + n = 63$ …(ii)

On solving eqs. (i) and (ii), we get

$n = 15$

and $x = 24$

27 Initially, the total age of family = $4 \times 42 + 7 \times 20 = 308$

and the average age = $\dfrac{308}{11} = 28$

Now, the total age of family

$$= 308 - (2 \times 13) + (11 + 15 + 28)$$
$$= 308 + 28 = 336$$

Now, the new average of the family = $\dfrac{336}{12} = 28$

Since, the average age of the original family and that of new family is same (*i.e.,* 28)

Hence, the average age of the new family is increased by 0 year.

28 Let there be n number and he missed a number k, then the

average (which he has calculated) = $\dfrac{\left(\dfrac{n(n+1)}{2} - k\right)}{n} = 15$

$\Rightarrow$ $n^2 - 29n = 2k$

$\Rightarrow$ $n(n - 29) = 2k$

Thus, at $n = 29$ or $n < 29$, the expression is invalid, since the value of k is neither zero nor negative, which is actually a natural number. So for the least possible valued of $n = 30$

$k = 15$

and for $n = 31$, $k = 31$

Again for $n > 31$, k is beyond the range *i. e.*, greater than n. Since, k cannot be greater than n. Hence, there are only two values of k. So there is no unique value of n.

29 Let there be n people (initially) in the group, then the total earning of the group = $n \times 50$

Again, $n \times 50 = (n - 2) \times 49 + (2x + 45)$

$\Rightarrow n = 2x - 53$; where x is the lowest earning of any person.

Now, since $42 < x < 47$ and $n \in$ prime numbers.

Then the only possible value of $n = 37$ for $x = 45$.

30 Let the number of students who got the jobs of A, B and C categories is a, b and c respectively,

then the total salary = $\dfrac{26(a + b) + 44(b + c) + 34(c + a)}{2(a + b + c)}$

$$= \frac{60a + 70b + 78c}{2(a + b + c)}$$
$$= \frac{30(a + b + c) + (5b + 9c)}{a + b + c}$$
$$= 30 + \text{some positive value}$$

So the minimum salary must be ₹ 30 lakh and the maximum salary cannot exceed 44, which is the highest of the three.

31 $D - C - A$ …(i)

$D > B > C$ …(ii)

From eqs. (i) and (ii), we get

$D > B > C > A$ …(iii)

Again, $E - B - A$

But $B > A$, from eq. (iii)

So, $E > D > B > C > A$ [Since, B is the average of E and A so it is equidistant from both E and A.]

32 Let Donald be denoted by H (Husband).

His wife be denoted by W (Wife).

His daughter be denoted by D (Daughter).

His son be denoted by S (Son).

The average age of 4 persons = $\dfrac{(H + W + D + S)}{4} = 23$

$\Rightarrow$ $H + W + D + S = 92$

Again, $H = W + 4$

	H		W	D	S
So, At the time when daughter is born	28	$\xleftarrow{(+4)}$	24	0	×
	↓				
At the time when son is born	32	$\xrightarrow{(-4)}$	28	4	0

So, at the time of birth of his son, total age of his family = 64 years (32 + 28 + 4 + 0 = 64)

and presently the total age of his family = 92 years

It means total increase in age of the whole family = 28 years

Thus the average increase in age $= \frac{28}{4} = 7$ years

It means the age of Donald = 39 years

and age of his daughter = 11 years

Therefore the average age of Donald and his daughter is 25 years.

33 Aman Wife Son El. D Yg. D

$5x$ x $5z$ $4z$

$8y$ $3y$

$\Rightarrow 40K$ W $8K$ $15K$ $12K$

Again since, Yg. $D + W = A + S$ $(K = x.y)$

$\Rightarrow 12K + W = 40K + 8K$ $W \rightarrow$ Age of wife

$\Rightarrow W = 36K$

Thus, 4 years ago $(36K - 4) = 8(8K - 4)$

$\Rightarrow 28K = 28 \Rightarrow K = 1$

Therefore, the age of Aman = 40

Wife = 36

Son = 8

Elder daughter = 15

Younger daughter = 12

Hence, the average age of the family $= \frac{111}{5} = 22.22$ years

34 Let there be n politicians (initially) in the party and their average weight be x kg, then $nx + 209 = \frac{5}{4} n(x - 1)$

$$\frac{n}{4}(x - 5) = 209$$

$$\Rightarrow \quad x = \frac{209 \times 4}{n} + 5$$

$$x = \frac{4 \times 11 \times 19}{n} + 5$$

So the possible value of n is 76 $(= 19 \times 4)$

Thus, $x = 16$

Therefore the average weight of all the politicians is 15 kg.

35 Total distance covered by all the wheels $= 4 \times 160$

Number of wheels used = 5

Therefore, average distance covered by each wheel

$$= \frac{4 \times 160}{5} = 128 \text{ km}$$

36 Let the numbers be

$(a - 5), (a - 3), (a - 1), (a + 1), (a + 3), (a + 5)$,

then their average

$$= \frac{(a-5)+(a-3)+(a-1)+(a+1)+(a+3)+(a+5)}{6} = a$$

Again the value of 'a' can be found by using the last statement

i. e., $(a - 5)^2 + (a + 5)^2 = 178$

$\Rightarrow \quad a^2 = 64 \Rightarrow a = 8$

37

	No. of Directors	Average Age	Total Age
Just before death and resignation	10	48	480
Just after death and resignation	9		{480 − (53 + x) + 34}
One year later	9	46	414

So, one year later, after the incident

total age $= \{480 - (53 + x) + 34\} + 9 \times 1 = 414$

$\Rightarrow \quad x = 56$ years

Where x is the age of the dead person at the time of his death.

38 Go through options $= \frac{24 \times 60 + 12 \times 54}{(24 + 12)}$

$= 58$, which is a whole number.

39

	Year/ Time	No. of Nurses	Average Age	Total Age
	1982	100	50	5000
Just before retirement	1984	100	52	5200
Just after retirement	1984	80	50	(5200 − 20 × 60) = 4000
Just before recruitment	1987	80	53	4240
Just after recruitment	1987	(80 + 40) = 120	48	(4240 + 38 × 40) = 5760
	1990	120	51	6120

40 Average score

$$= \frac{8+10+7+12+25+24+18+23+29+30}{10} = \frac{186}{10} = 18.6$$

41 The candidate who has scored maximum (means continuing for long time and number of responses were maximum) can be such a required person.

Now, consider Abhijeet, whose score was 30.

Again consider Rajesh, Radhe, Harish and one more candidate

Since, $8 + 10 + 7 < 30 \Rightarrow 25 < 30$

So there are exactly 3 persons which can be full time opponents.

Further there are 5 (= 30 – 25) more chances. So these chances can be utilised by any other candidate but he or she must be in the beginning or in the ending of the Abhijeet, since the score of all the candidates is greater than 5. Hence there can be maximum 3 + 1 = 4 opponents of Abhijeet.

42 There is no such an information.

43 The minimum number of participants who have lost the game before Abhijeet started will be possible only when the number of participant with Abhijeet be maximum, which is 4.

Hence, excluding these 4 players (opponents of Abhijeet) and Abhijeet himself there are 5 people left. So, minimum 5 players have lost the game (or contest).

44 The opponents of Abhijeet (as per the requirement) are Rajesh, Radhe, Harish and Amit Sana

So, the average score $= \frac{8 + 10 + 7 + 29}{4} = 13.5$

45 Go through options or total employees $= \frac{(25 - 10) \times 20}{10} = 30$

Hence, number of educated employees = 30 – 20 = 10

46 Average speed of the later half journey

$$= \frac{2 \times 40 \times 60}{100} = 48 \text{ km/hr}$$

Now the average speed of the whole journey

$$= \frac{2 \times 48 \times 96}{144} = 64 \text{ km/hr}$$

47 $\frac{40 \times 0.8 + 60 \times 0.75 + 50 \times 0.6 + 30 \times 0.5}{180} \times 100$

$$= \frac{122}{180} \times 100 = 67.777 \ldots \%$$

48 Remember $\begin{array}{r} abc \\ -\,cba \\ \hline 99\,(a - c) \end{array}$

where abc and cba are the three digit numbers and $(a, c) \neq 0$

Again, since the difference in average = 19.8

Therefore the difference in total = 19.8 × 10 = 198

Thus, $99 \times (a - c) = 198$

$\therefore \quad (a - c) = 2$

49 If the highest weight be 35 gm, then the second highest weight will be 31 gm. Again if the second highest will be 31, then the third highest will be 33 which is in admissible, since then 35 + 33 = 68 which is not the greatest possible combination. Hence, wrong.

Similarly, 36 (*i. e.*, option *b*) is also invalid

Highest	Sec. Highest	Third Highest
36	30	34

Thus, 36 + 34 = 70 > 66, hence wrong.

The greatest possible combination cannot be greater than 66.

Now, consider option (c)

Highest	Sec. Highest	Third Highest	
34	32	32	× (since, weights are different)
	32	31	× (since, 65 is not a combination)
	32	30	✓

So, the highest weight 34

Sec. highest weight 32

Third highest weight 30

Lowest weight 29

Since, all the weights obtained give all the 6 different combinations, hence 34 is the highest possible weight of an envelope.

50 Let the fixed expenditure of the hotel be ₹ x and the variable expenditure (which is dependent on the guest) is ₹ y, then

$$x + 10y = 600 \quad \ldots(i)$$

$$x + 20y = 800 \quad \ldots(ii)$$

$$\Rightarrow \quad 10y = 200$$

$$y = ₹\,20 \quad \text{and} \quad x = 400$$

Hence the total expenditure when there are 40 guests

$$= 400 + 40 \times 20 = 1200$$

Therefore, average expenditure $= \frac{1200}{40} = ₹\,30$

1 Since all the total 100 elements of Sets A, B, C are the natural numbers upto. Thus the average of these first 100 natural numbers is the required average.

$\therefore$ Average $= \frac{1 + 2 + 3 + 4 + \ldots + 100}{100} = \frac{100 \times 101}{2 \times 100} = 50.50$

2 Except to 2 there are all the even numbers upto 100

So, the required average $= \frac{(2 + 4 + 6 + \ldots + 100) - 2}{49}$

$$= \frac{50 \times 51 - 2}{49} = \frac{2548}{49} = 52$$

NOTE There are only 49 elements in the Set B. Apply the formula of sum of first even numbers. Also use the property of AP.

3 The total value of all the 25 elements of the Set A

$$= 25 \times 42.4 = 1060$$

Since, there are 25 prime numbers upto 100 in the Set A

Again in the Set A and C there are 50 odd numbers and one even number. So the sum of all the elements of A and C

$$= (1 + 3 + 5 + 7 + \ldots + 99) + 2 = (50)^2 + 2 = 2502$$

Therefore the sum of all the elements of Set C

$$= 2502 - 1060 = 1442$$

Hence, the average of the Set C

$$= \frac{1442}{26} = 55.4615$$